Predicting Customer Churn in Telecom

Mehdi Chtourou, Jean Courtial, Thomas Pitié, Carl Claise

What is Churn ?

Customer churn refers to the event where a customer stops using a company’s product or service during a given time period.

Why Does Churn Matter?

- Customer acquisition costs about 5-7x more than retention
- Churn directly impacts: revenue, growth and customer life value
- High churn often signals pricing, service quality, or customer experience issues

Business Context: Why Predict Churn?

Early detection allows:

- Targeted retention offers (Price)
- Personalized communication (Service Quality)
- Improved customer experience (Customer Experience)

Goal:

- Use data to identify at-risk customers early and proactively reduce cancellations

Problem Statement: Customer Churn

Business Question:

- Which customers are most likely to churn, and why ?

Analytical Objectives:

- Predict the probability of customer churn
- Identify the key drivers behind churn behavior
- Translate model insights into actionable business decisions

Dataset Overview

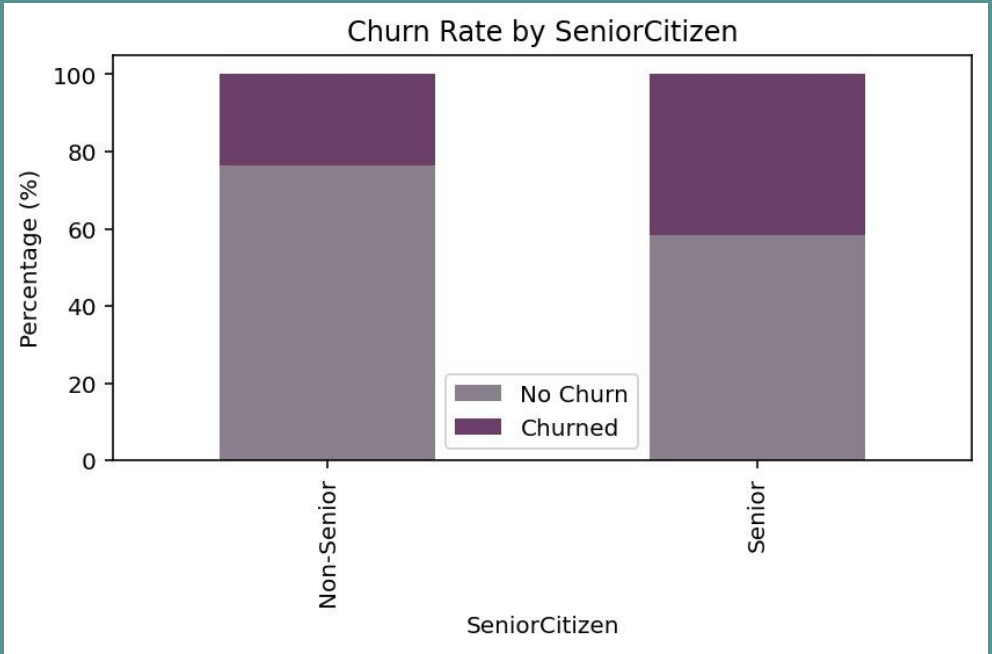
Telco Customer Churn Dataset (Kaggle)

- 7,000 Customers
- Target Variable: Churn = Customer left the company last month

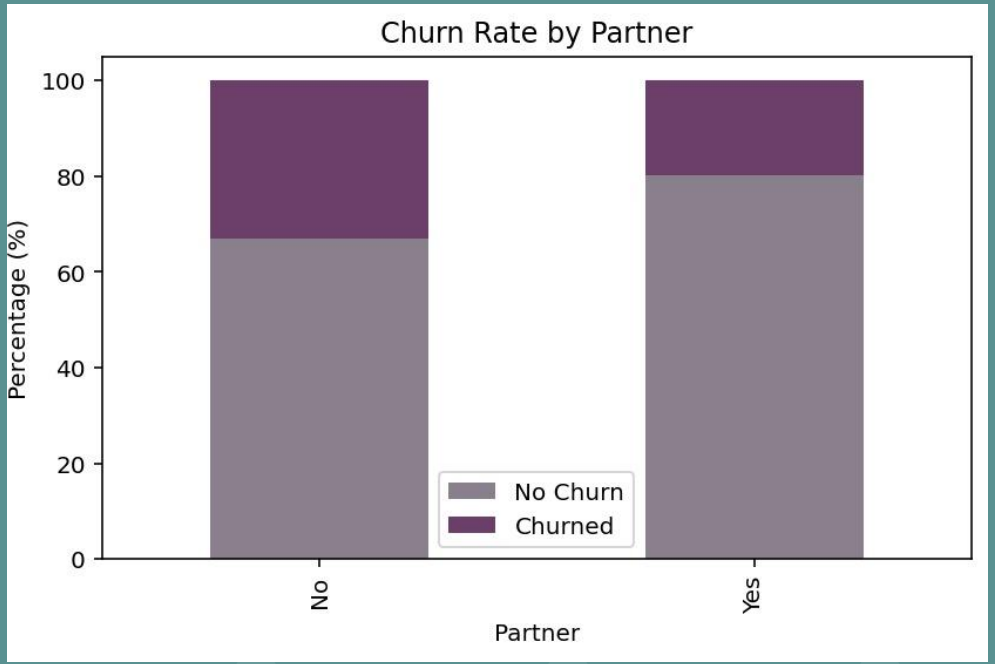
Features:

- Demographics (age proxy, senior citizen, sex)
- Account info (tenure, contract)
- Services (internet, phone, streaming)
- Billing (monthly, total charges)

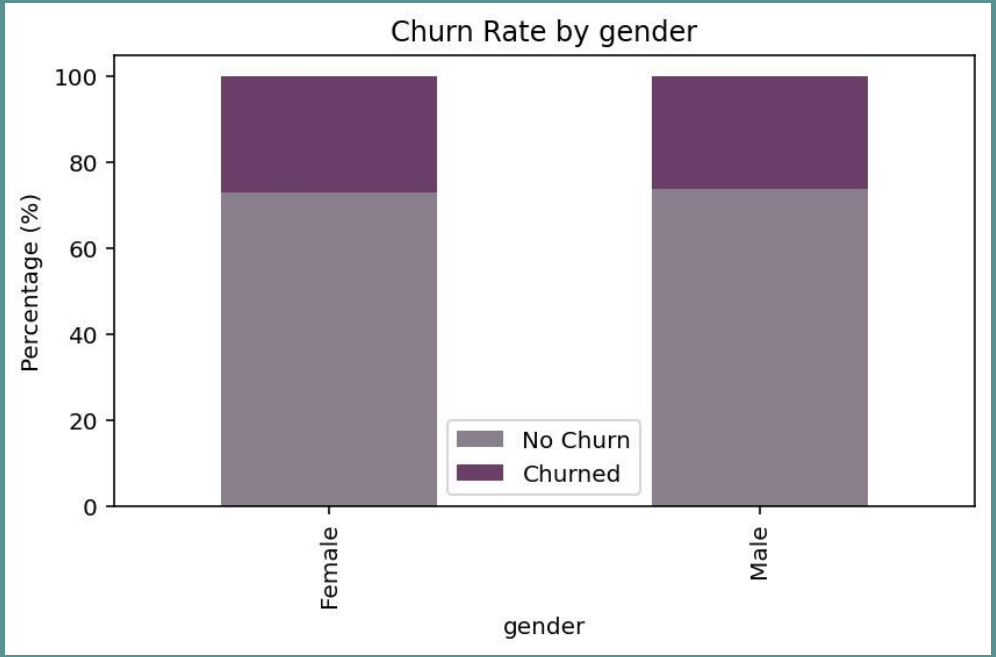
Which Customers Churn the Most?



Which Customers Churn the Most?



Which Customers Churn the Most?



Two Approaches

- Using RandomForestClassifier: one model takes into account all parameters, the second takes the best five from the first

- Using xGBoost

RandomForest - Results Overview

- About 26.6% of customers churn
- Churn recall of 77% → very satisfying
- ROC AUC of 83% meaning we have a strong model
- Most important features :
  - Contract: 47%
  - Online Security: 17%
  - Tech Support: 15%
  - Internet Service: 11%
  - Payment Method: 9%

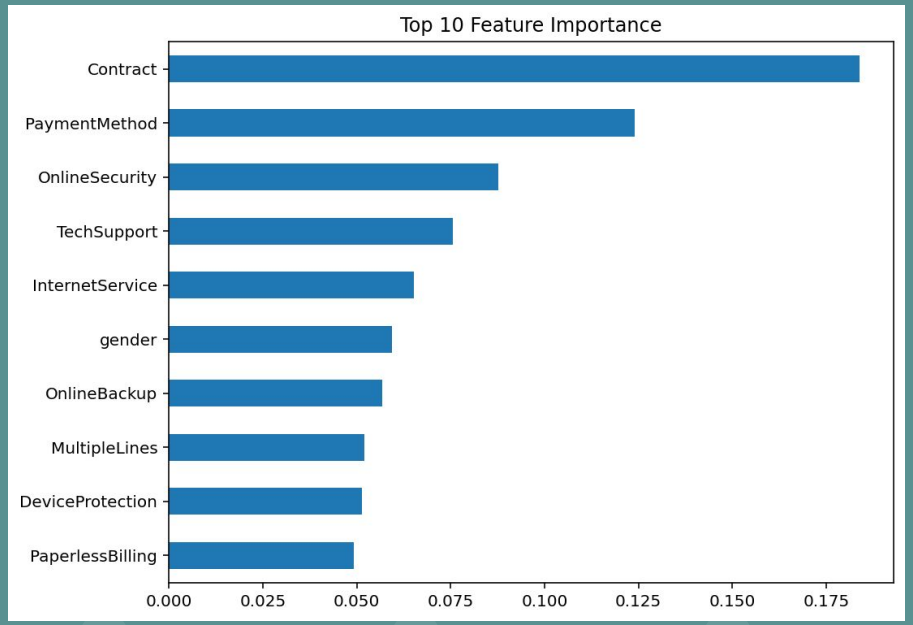
xGBoost- Results Overview

- About 26.6% of customers churn
- Churn recall of 85% → very good
- ROC AUC of 83% meaning we have a strong model
- Precision: 45% which is average for churn models
- Most important features :
  - Contract: 40%
  - Online Security: 17%
  - Tech Support: 15%
  - Internet Service: 8%
  - Payment Method: 4%

What drives churn ?

Key Drivers

- Contract type is the strongest predictor of churn
- Short tenure customers are more likely to leave
- Higher monthly charges increase churn risk
- Online security and tech support reduce churn



Customer Lifecycle & Pricing Effects

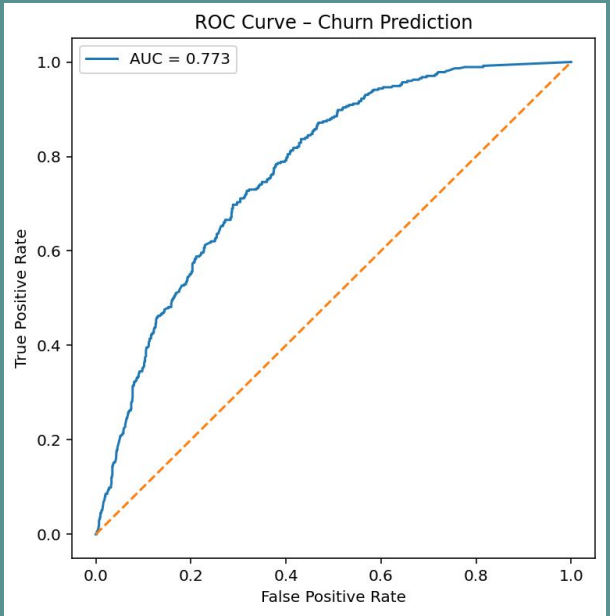
Lifecycle Insights

- Tenure is strongly negatively correlated with churn (−0.35)
- Customers are most at risk during the early months
- Higher monthly charges slightly increase churn
- Total charges increase with tenure, reflecting customer lifetime value

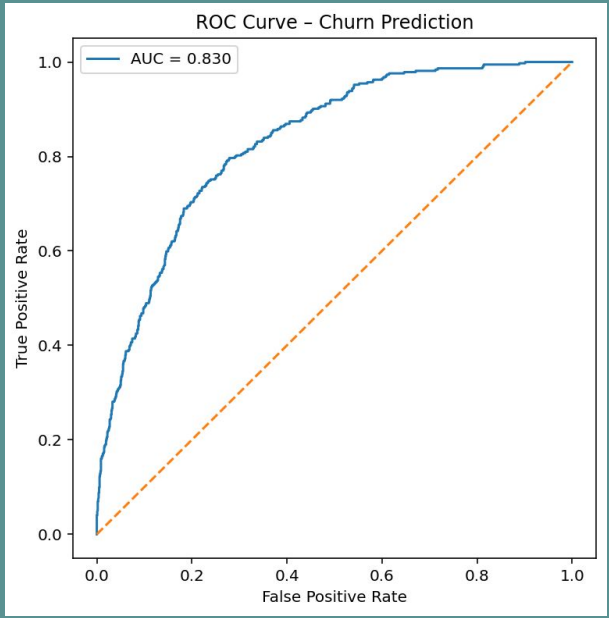


ROC Curve

RandomForestClassifier



xGBoost

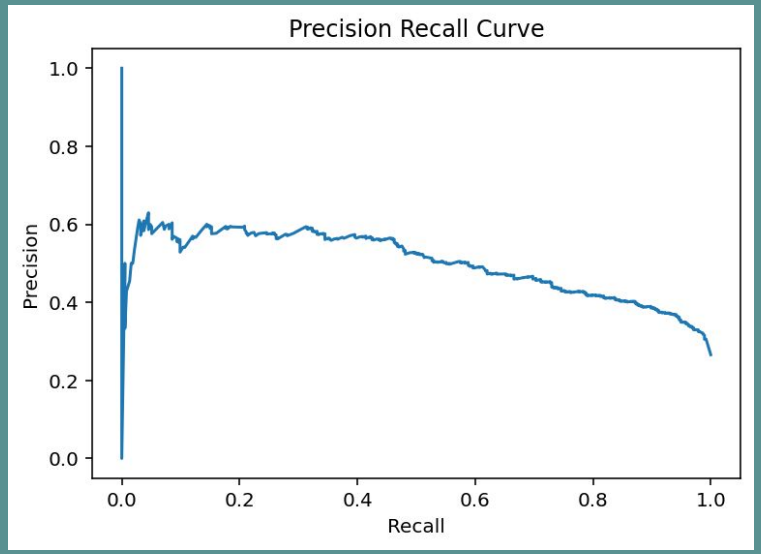


In both cases, ROC AUC > 0.77, indicating strong model performance

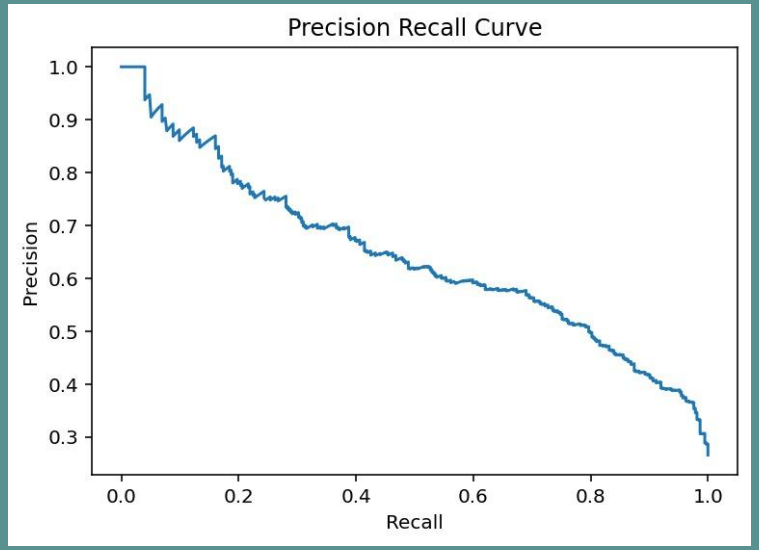
The model effectively distinguishes churn vs. non-churn customers

Precision Recall Curve

RandomForestClassifier

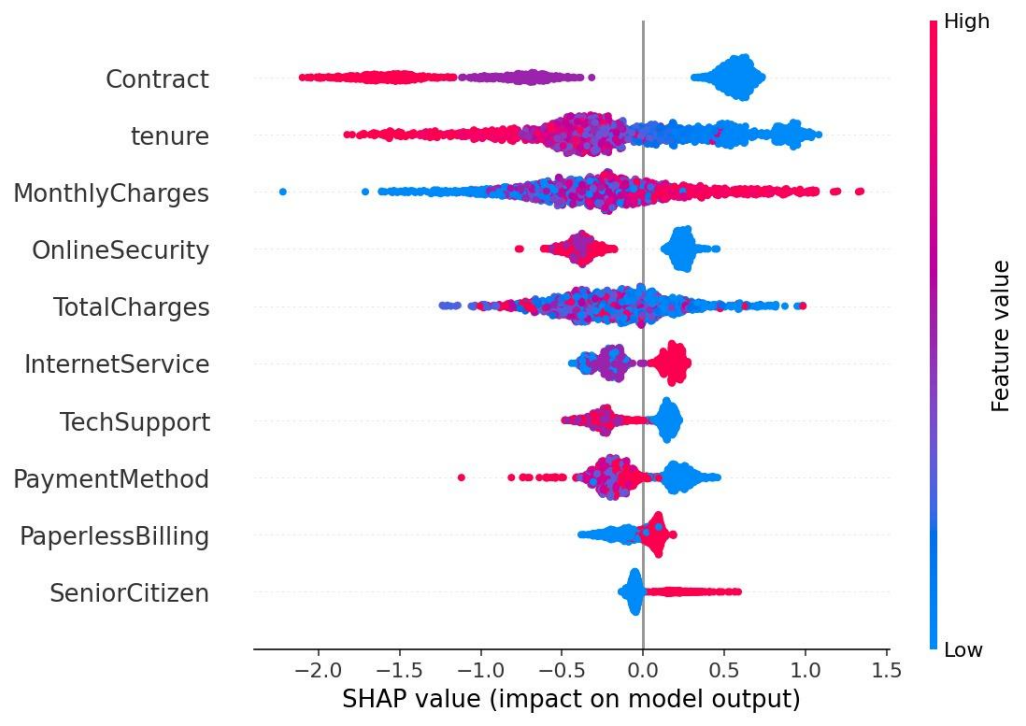


xGBoost



Precision Recall results support reliable targeting of high-risk customers

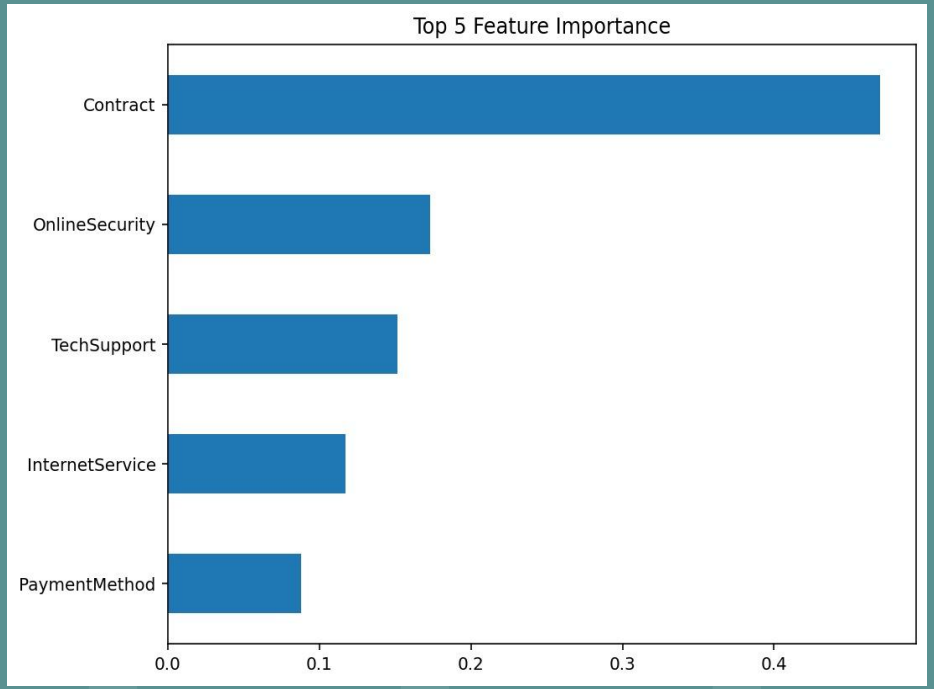
SHAP For Explainability



Key Business Levers

Main Action Areas

- Encourage customers to switch to long-term contracts
- Improve value perception for high-price plans
- Promote security and technical support services
- Focus retention efforts on new customers



Final Business Recommendations

How to Reduce Churn

- Target new customers during the first 3–6 months
- Offer incentives for long-term contracts
- Identify high-risk customers using the model
- Provide personalized retention offers (bundles, discounts, support)

Impact

- Proactive retention strategy
- Better customer lifetime value
- Reduced revenue loss